12
Long Process Dead-Time in Closed Loop Control and the Smith Predictor

12.1 Objectives

As a result of studying this chapter, and after having completed the relevant exercises, the student should be able to:

- Demonstrate the correct use of a process simulation for process variable prediction;
- Show how control loops with long dead-times are dealt with correctly;
- List the procedures for tuning of control loops with long dead-times.

12.2 Process Deadtime

Overcoming the dead time in a feedback control loop can present one of the most difficult problems to the designer of a control system. This is especially true if the dead time is > than 20% of the total time taken for the PV to settle to its new value after a change to the SP value of a system.

We have seen that little or no dead time in a control system presents us with a simple and easy set of algorithms that when applied correctly give us extremely stable loop characteristics.

Unfortunately, if the time from a change in the manipulated variable (Controller Output) and a detected change in the PV occurs, is excessive, any attempt to manipulate the process variable before the dead time has elapsed will inevitably cause unstable operation of the control loop.

Figure 12.1 illustrates various dead times and their relationship to the PV reaction time.

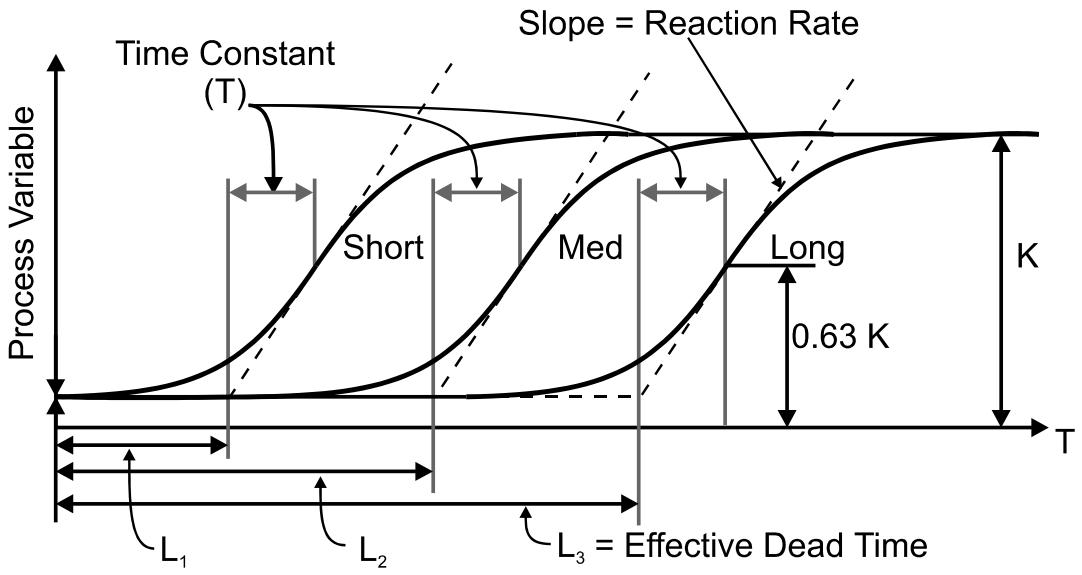


Figure 12.1
Reaction Curves showing short, medium and long dead times

12.3 An Example of Process Deadtime

Process Deadtime occurs in virtually all types of process, as a result of the PV measurement being some distance away, both physically and in time, from the actuator that is driven by the manipulated variable.

An example of this is in the overland transportation of material from a loading hopper to a final process, this being some distance away. The critical part of the operation is to detect the amount of material arriving at the end of its journey, the end of the conveyor belt, and from this performing two functions:

- To "tell" the ongoing process how much material is arriving and
- To adjust the hopper feed rate at the other end of the belt.

Figure 12.2 illustrates this problem, the controller is measuring the weight of arriving material that during its journey from the supply hopper has encountered some loss due to spillage from the conveyor. Also the amount of material deposited on the belt has varied due to variability of the amount, or head, of material in the hopper.

The Deadtime can be calculated very simply by the product of the belt speed and the distance between the input hopper; where the action of the manipulated variable (Controller Output) occurs and the PV or point where the belt weigher is located.

In this example, the controller measures the weight / meter / minute of the arriving material, compares this with the SP and generates an output, but now it must wait for the Deadtime period, which in this example is about 10 minutes, before seeing a result of this change in the value of the MV. If the controller expects a result before the Deadtime has elapsed, and none occurs, it will assume that its last change had no effect and it will continue to increase its output until such time as the PV senses a change has occurred. By this time it will be too late, the controller will have overcompensated, either by now supplying too much or too little material.

The magnitude of this resultant error will depend on the sensitivity of the system and the difference between the assumed and actual Deadtime. That is, if the system is highly sensitive (high gains and fast responses tuned into it) it will effect large movements of the inlet hopper for small PV changes. Also if the assumed Deadtime is much shorter than the actual Deadtime it will spend longer changing its output (MV) before sensing a change in the PV.

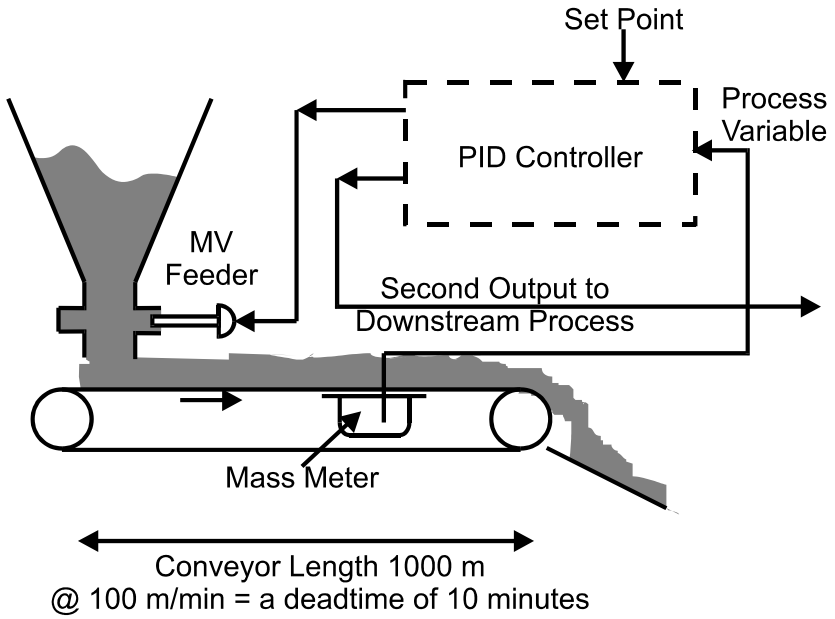


Figure 12.2
Illustration of a long conveyor system giving an excessive deadtime to the control loop

12.3.1 Overcoming Process Deadtime

To cure these problems depends; to a great extent; on the operating requirement(s) of the process. The easiest solution is to "de-tune" the controller to a slower response rate. The controller will then not overcompensate unless the dead time is excessively long.
The integrator (I mode) of the controller is very sensitive to "Deadtime" as during this period of inactivity of the PV (an ERR term is present) the integrator is busy "ramping" the output value.

Ziegler and Nichols determined the best way to "de-tune" a controller to handle a Deadtime of D minutes is to reduce the Integral time constant T_{INT} by a factor of D^2 and the Proportional constant by a factor of D.

The derivative time constant T_{DER} is unaffected by dead time as it only occurs after the PV starts to move.

If, however, we could inform the controller of the Deadtime period, and give it the patience to wait and be content until the Deadtime has passed then detuning and making the whole process very sluggish would not be required.

This is what the Smith Predictor attempts to perform.

12.4 The Smith Predictor Model

In 1957 O.J.M. Smith, of the University of California at Berkeley proposed the predictor control strategy as explained below. Figure 12.3 illustrates the mathematical model of the predictor which consists of:

- An ordinary feedback loop;
- A second, or inner, loop that introduces two extra terms into the feedback path.

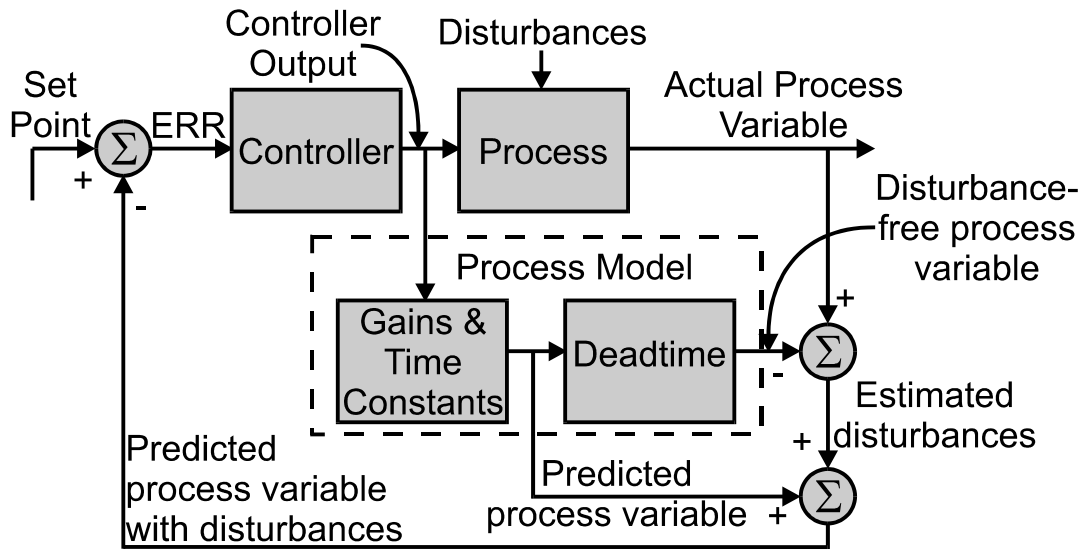


Figure 12.3
The Smith Predictor Model

12.4.1 First Term Explanation (Disturbance Free PV)

The first term is an **ESTIMATE** of what the **PV** would be like in the absence of any process disturbances. It is produced by running the controller output through a model that is designed to accurately represent the behavior of the process without taking any load disturbances into account. This model consists of two elements connected in series.

1. The first represents all of the process behavior not attributable to Deadtime. This is usually calculated as an ordinary differential or difference equation that includes estimates of all the process gains and time constants.
2. The second represents nothing but the Deadtime and consists simply of a time delay, what goes in, comes out later, unchanged.

12.4.2 Second Term Explanation (Predicted PV)

The second term introduced into the feedback path is an estimate of what the PV would look like in the absence of both disturbances and Deadtime.
It is generated by running the controller output through the first element of the model (Gains and TC's) but NOT through the time delay element.

It thus ***Predicts*** what the disturbance-free PV will be like once the Deadtime has elapsed.

12.5 The Smith Predictor in Theoretical Use

Figure 12.4 shows the Smith Predictor in a practical configuration, or as it is really used.

It shows an estimate of the PV (with both disturbances and dead time) generated by adding the estimated disturbances back into the disturbance-free PV. The result is a feedback control system with the deadtime outside the loop.

The Smith predictor essentially works to control the modified feedback variable (The predicted PV with disturbances included) rather than the actual PV.

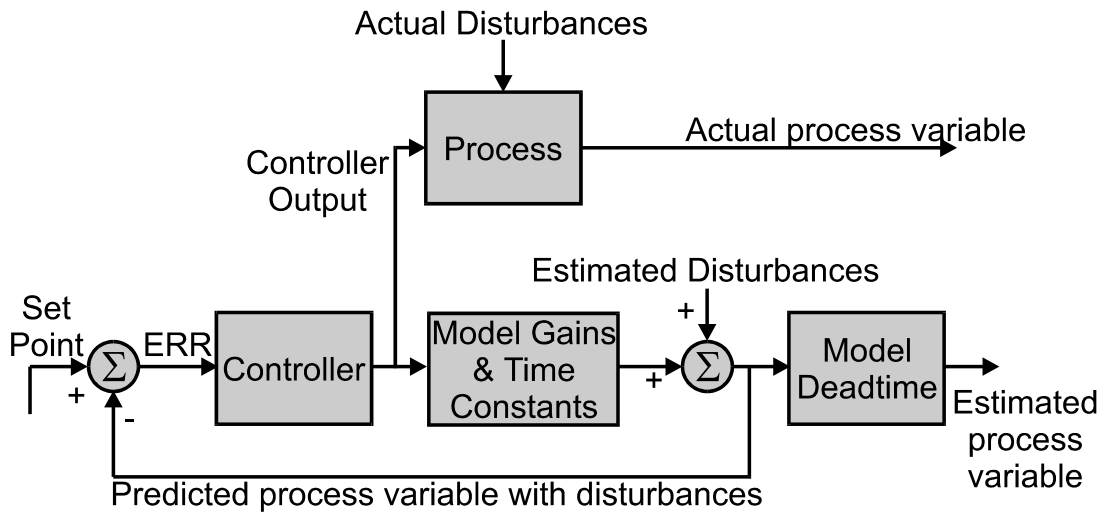


Figure 12.4
The Smith Predictor in Use

If it is successful in doing so; and the process model accurately emulates the process itself, then the controller will simultaneously drive the actual PV toward the SP value, irrespective of SP changes or load disturbances.

12.6 The Smith Predictor in Reality

In reality there is plenty of room for errors to creep into this "Predictive Ideal". The slightest mismatch between the process dynamic values and the model can cause the controller to generate an output that successfully manipulates the modified feedback variable but drives the actual PV into nihilism, never to return.

There are many variations on the Smith predictor principle, but readmits, especially long ones, remain a particularly difficult control problem to control and solve.

12.7 An Exercise in Deadtime Compensation

We have seen that if a long dead-time is part of the process behavior, the quality of control becomes unacceptably low. The main problem lies in the fact that the reaction to a MV change is not seen by the PV until the Deadtime has expired. During this time, neither a human operator nor an automatic controller knows how the MV change has effected the process.

Exercise Fourteen illustrates the concepts of Deadtime compensation, based on the arrangement shown in figure 12.5.

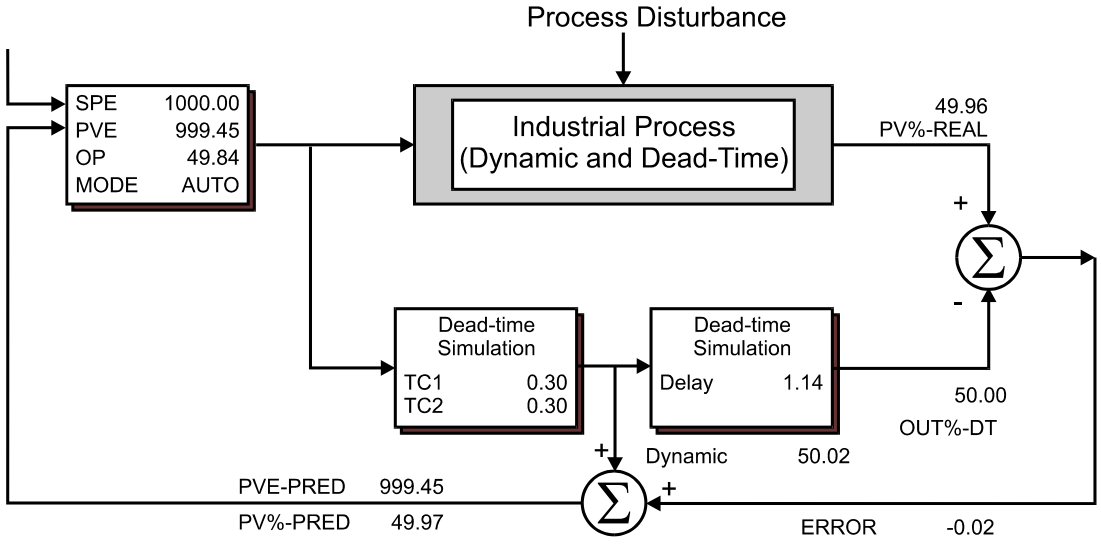


Figure 12.5
Block Diagram of Exercise 14; Closed Loop Control with Process Simulation

As there is no means of separating process dead-time from process dynamic in order to find out how the process would behave without dead-time, we make use of the values provided by a process simulation.

The process simulation is split into two parts as seen in Figure 12.5 these two parts are described in sections 12.4.1 and 12.4.2.